

NADA MOHAMED

AI/ML Engineer

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Summary

Computer Science graduate (2026) specializing in Artificial Intelligence and Machine Learning, with an engineering background spanning intelligent agents, LLM fine-tuning, computer vision, full-stack development, blockchain, and fintech. Shipped 15+ projects end to end and taught programming to 30+ students, equally at home building systems and explaining them.

Education

Arab Academy for Science, Technology & Maritime Transport **Sep 2022 – Jun 2026**
B.Sc. Computer Science (Dual Degree, University of Northampton UK). Honours, GPA 3.69 / 4.00 Cairo, Egypt

Skills

Languages: Python, Go, Java, C/C++, JavaScript, TypeScript, PHP, Dart, R, Bash, SQL, HTML/CSS

AI / ML & Data: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV, Librosa, HuggingFace, Whisper; NLP, Generative AI & LLM fine-tuning, Computer Vision, Reinforcement Learning, Explainable AI (SHAP, Grad-CAM)

Web & Application: Flask, FastAPI, React, Tailwind CSS, Vite, JavaFX, Tkinter, Flutter, SwiftUI, PyWebView

Systems, Cloud & DevOps: Docker, Kubernetes, Git, GitHub Actions (CI/CD), MySQL, AWS RDS, SQLite; parallel computing, compilers, Arduino

Blockchain & Security: Hyperledger Fabric, Chaincode (Go), X.509 PKI, cryptography, threshold (MPC) custody

Soft Skills: Mentorship & instruction, team collaboration, technical writing & documentation, analytical thinking

Experience

SAIB Bank **Aug 2026 – Present**
Fintech & Startup Partnerships Intern Mohandeseen, Cairo

- Working under the Head of Fintech & Startup Partnerships on applying emerging technology to bank operations.
- Authoring a business case for a chosen product, applying business-cycle analysis end to end.

Dell Technologies **Aug 2025 – Sep 2025**
Summer Intern Cairo, Egypt

- Led the Academy Project end to end: scoped the idea, built the case, presented the proposal.

DEPI – Microsoft ML Engineering Curriculum **Nov 2024 – May 2025**
AI/ML Engineer Intern Zamalek, Egypt

- Completed a 7-month intensive AI/ML program covering NLP, generative AI, computer vision, and deployment.
- Delivered milestone projects and a capstone integrating multimodal workflows and agent orchestration.

Megz (IGCSE Computer Science) **Aug 2020 – Nov 2021**
Computer Science Instructor Cairo, Egypt

- Taught Python and Java fundamentals to 30+ students across multiple IG schools.
- Supported the lead teacher in classroom management, grading, and student progress tracking.
- Mentored students in debugging and problem-solving; personalized guidance improved small-group performance 20%.

Projects

Payramid: Blockchain Payroll for Egypt 🌐🔗 **Sep 2025 – Present**
Go, TypeScript, Blockchain, Docker · Graduation Project

- Payroll orchestration and immutable audit layer letting employers disburse salaries over Egypt's national payment rails with verifiable proofs of payment, within local banking and data-protection regulation.
- Authored the thesis and the IEEE 830 SRS start to finish; currently revising the accompanying research paper.

Music Genre Detection 🔗 **Dec 2025**
Python, PyTorch, Librosa, Flask, Captum

- CNN-LSTM attention model classifying audio into 10 genres from mel-spectrograms, served through Flask.
- Grad-CAM and Integrated Gradients explainability, a pytest suite with coverage, and Sphinx API docs.

Pixel Paradise 🌐 **May 2024 – Feb 2025**
JavaScript, PHP, HTML/CSS

- Lead contributor on a four-person team building a browser arcade platform: built four of the seven games end to end (Pong with single-player and real-time two-player modes, Tetris, Snake, and Mario Bros), plus the site shell and front-end rendering, wiring game scores and session state into the shared leaderboard.

Certifications, Awards & Languages

🏆 **First Place, AI/ML Project Submission – AASTMT** • **Microsoft ML Engineer Course – DEPI**

🏆 **AI & Machine Learning Course – Universitat Autònoma de Barcelona** • **CS Camp – Carnegie Mellon University**

Languages: Arabic (Native) • English (Fluent) • Spanish (Intermediate)